

10/16/2024 Proof of NC = SC for Convex Function

Convex function.

Sufficient Condition = Necessary Condition

if you find a point

$$\begin{cases} \nabla f(x^*) = 0 \\ \nabla^2 f(x^*) \succeq 0 \end{cases}$$

$\Rightarrow$ local min
||
global min

$$\nabla f(x^*) = 0 \Rightarrow \text{global min.}$$

Suppose x^* satisfies $\nabla f(x^*) = 0$

Invoke second-order MVT on any given x

$$f(x) = f(x^*) + \langle \nabla f(x^*), x - x^* \rangle + \frac{1}{2} (x - x^*)^T \nabla^2 f(\bar{x}) (x - x^*)$$

(for some point) $\bar{x}$ in between x and x^*

$$= f(x^*) + \frac{1}{2} (x - x^*)^T \nabla^2 f(\bar{x}) (x - x^*)$$

$\succeq 0$

≥ 0

$\geq f(x^*)$. That is, $f(x) \geq f(x^*), \forall x$.